Programming Fundamentals

WEEK #3

Arab Academy
for Science, Technology and Maritime Transport

Lecture Agenda

- The parts of a C++ Program
- The cout object
- Variables and Literals
- Data types
- The cin object

The Parts of a C++ Program

The // marks the beginning of a comment

Program 2-1

```
1  // A simple C++ program
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7     cout << "Programming is great fun!";
8     return 0;
9  }
```

this line starts with a #, it is called a preprocessor directive

This marks the beginning of a function. A function can be thought of as a group of one or more programming statements that collectively has a name

The main function must start with { and end with }

The output of the program is shown below. This is what appears on the screen when the program runs.

Program Output

```
Programming is great fun!
```

The main function ends with return 0

Every statement must end with a semi colon

The cout object

- Use the cout object to display information on the computer's screen.

Program 2-2

```
// A simple C++ program
#include <iostream>
using namespace std;

int main()
{
   cout << "Programming is " << "great fun!";
   return 0;
}
```

Program Output

```
Programming is great fun!
```

Program 2-3

```
// A simple C++ program
#include <iostream>
using namespace std;

int main()
{
   cout << "Programming is ";
   cout << "great fun!";
   return 0;
}
```

Program Output

```
Programming is great fun!
```

Program 2-4

```
// An unruly printing program
#include <iostream>
using namespace std;

int main()
{
   cout << "The following items were top sellers";
   cout << "during the month of June:";
   cout << "Computer games";
   cout << "Coffee";
   cout << "Aspirin";
   return 0;
}
```

Program Output

```
The following items were top sellersduring the month of June:Computer gamesCoff
eeAspirin
```

The cout object

Program 2-5

```
// A well-adjusted printing program
#include <iostream>
using namespace std;

int main()
{
   cout << "The following items were top sellers" << endl;
   cout << "during the month of June:" << endl;
   cout << "Computer games" << endl;
   cout << "Coffee" << endl;
   cout << "Aspirin" << endl;
   return 0;
}
```

Program Output

```
The following items were top sellers
during the month of June:
Computer games
Coffee
Aspirin
```

Program 2-6

```
// Yet another well-adjusted printing program
#include <iostream>
using namespace std;

int main()
{
   cout << "The following items were top sellers\n";
   cout << "during the month of June:\n";
   cout << "Computer games\nCoffee";
   cout << "\nAspirin\n";
   return 0;
}
```

Program Output

```
The following items were top sellers
during the month of June:
Computer games
Coffee
Aspirin
```

Variables and Literals

Variables represent storage locations in the computer's memory. Literals are constant values that are assigned to variables.

This is called a variable definition.
int is the data type
number is the variable name

Program 2-7

```
// This program has a variable.
#include <iostream>
using namespace std;

int main()
{
   int number;

   number = 5;
   cout << "The value in number is " << number << endl;
   return 0;
}
```

This is called an assignment.

Program Output

```
The value in number is 5
```

Variables and Literals

Rules that must be followed with all identifiers.

- The first character must be one of the letters a through z, A through Z, or an underscore character (_).
- After the first character you may use the letters a through z or A through Z, the digits 0 through 9, or underscores.
- Uppercase and lowercase characters are distinct. This means ItemsOrdered is not the same as itemsordered.i.e C++ is case sensitive.

Variables and Literals

Table 2-5 Some Variable Names

Variable Name	Legal or Illegal?
`dayOfWeek`	Legal.
`3dGraph`	Illegal. Variable names cannot begin with a digit.
`_employee_num`	Legal.
`June1997`	Legal.
`Mixture#3`	Illegal. Variable names may only use letters, digits, or underscores.

Data Types

Data Type	Declaration	Example
Integers	int	3,5,123,-589
Decimal	float	3.5, -45.8, 20.0
Characters	char	‘a’ ,’B’, ‘5’, ‘ *’

Data Types

Program 2-11

```
1  // This program shows three variables defined on the same line.
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7     int floors, rooms, suites;
8
9     floors = 15;
10    rooms = 300;
11    suites = 30;
12    cout << "The Grande Hotel has " << floors << " floors\n";
13    cout << "with " << rooms << " rooms and " << suites;
14    cout << " suites.\n";
15    return 0;
16 }
```

Program Output

```
The Grande Hotel has 15 floors
with 300 rooms and 30 suites.
```

Program 2-13

```
1  // This program uses character literals.
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7     char letter;
8
9     letter = 'A';
10    cout << letter << endl;
11    letter = 'B';
12    cout << letter << endl;
13    return 0;
14 }
```

Program Output

```
A
B
```

Program 2-15

```
1  // This program uses floating point data types.
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7     float distance;
8     double mass;
9
10    distance = 1.495979E11;
11    mass = 1.989E30;
12    cout << "The Sun is " << distance << " meters away.\n";
13    cout << "The Sun\'s mass is " << mass << " kilograms.\n";
14    return 0;
15 }
```

Program Output

```
The Sun is 1.49598e+011 meters away.
The Sun's mass is 1.989e+030 kilograms.
```

The cin object

The cin object can be used to read data typed at the keyboard.

Program 3-1

```
// This program asks the user to enter the length and width of
// a rectangle. It calculates the rectangle's area and displays
// the value on the screen.
#include <iostream>
using namespace std;

int main()
{
   int length, width, area;

   cout << "This program calculates the area of a ";
   cout << "rectangle.\n";
   cout << "What is the length of the rectangle? ";
   cin >> length;
   cout << "What is the width of the rectangle? ";
   cin >> width;
   area = length * width;
   cout << "The area of the rectangle is " << area << ".\n";
   return 0;
}
```

Program Output with Example Input Shown in Bold

```
This program calculates the area of a rectangle.
What is the length of the rectangle? 10 [Enter]
What is the width of the rectangle? 20 [Enter]
The area of the rectangle is 200.
```

Program 3-2

```
// This program asks the user to enter the length and width of
// a rectangle. It calculates the rectangle's area and displays
// the value on the screen.
#include <iostream>
using namespace std;

int main()
{
   int length, width, area;

   cout << "This program calculates the area of a ";
   cout << "rectangle.\n";
   cout << "Enter the length and width of the rectangle ";
   cout << "separated by a space.\n";
   cin >> length >> width;
   area = length * width;
   cout << "The area of the rectangle is " << area << endl;
   return 0;
}
```

Program Output with Example Input Shown in Bold

```
This program calculates the area of a rectangle.
Enter the length and width of the rectangle separated by a space.
10 20 [Enter]
The area of the rectangle is 200
```

Exercise :

Programming Challenges

a) Write a program that stores the integers 62 and 99 in variables, and stores the sum of these two in a variable named total.

b) Repeat the above program but allow the user to enter the two integers x and y and then stores their sum in a variable named total.

c) Write a program that will compute the total sales tax on a $52 purchase. Assume the state sales tax is 4 percent and the county sales tax is 2 percent

Thank You